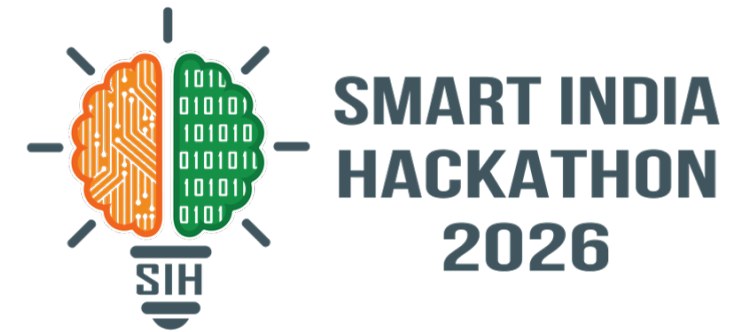




SMART INDIA HACKATHON 2026



- **PROBLEM STATEMENT ID – SIH26104**
- **PROBLEM STATEMENT TITLE- AI-POWERED REAL-TIME DETECTION AND PREVENTION OF VOICE CLONING IMPERSONATION ATTACKS**
- **THEME- MISCELLANEOUS**
- **PS CATEGORY- SOFTWARE**
- **TEAM ID-**
- **TEAM NAME- VERITY**



TITLE PAGE

INCOMING CALL

Caller : Brother (Claimed)

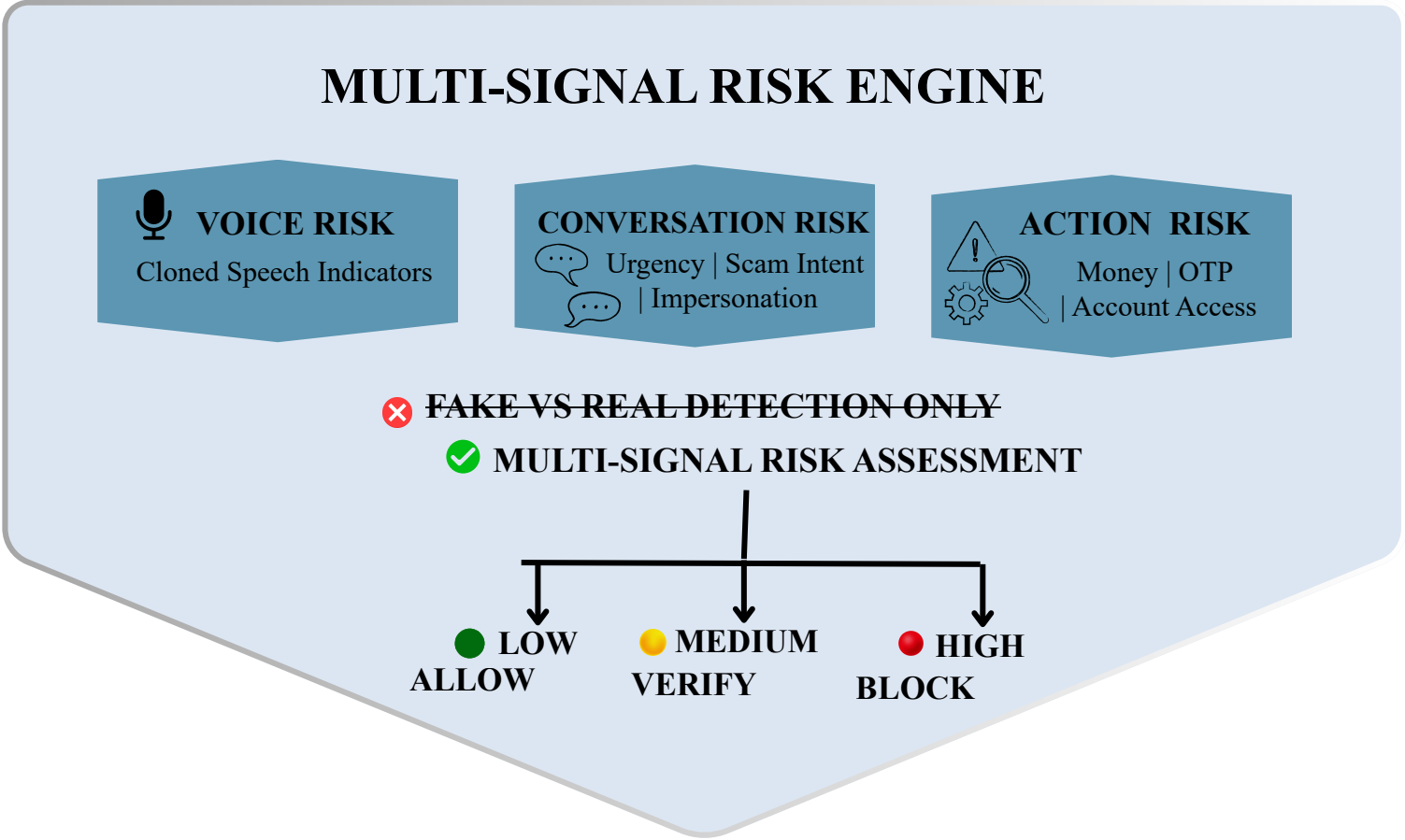
"I'm your brother.
I urgently need ₹25,000."

Urgency Signal

Financial Request

Impersonation Claim

Potential Attack Vector Detected



Higher Consequence => Stronger Verification

INDEPENDENT VERIFICATION

TrustBridge Alert

A Caller Claiming to be your brother requested :

₹25,000

Did you make this request ?

Yes - Approved

No - Blocked

Fraud Prevented — Suspicious interaction blocked before authorization

WHY TRUSTBRIDGE ??

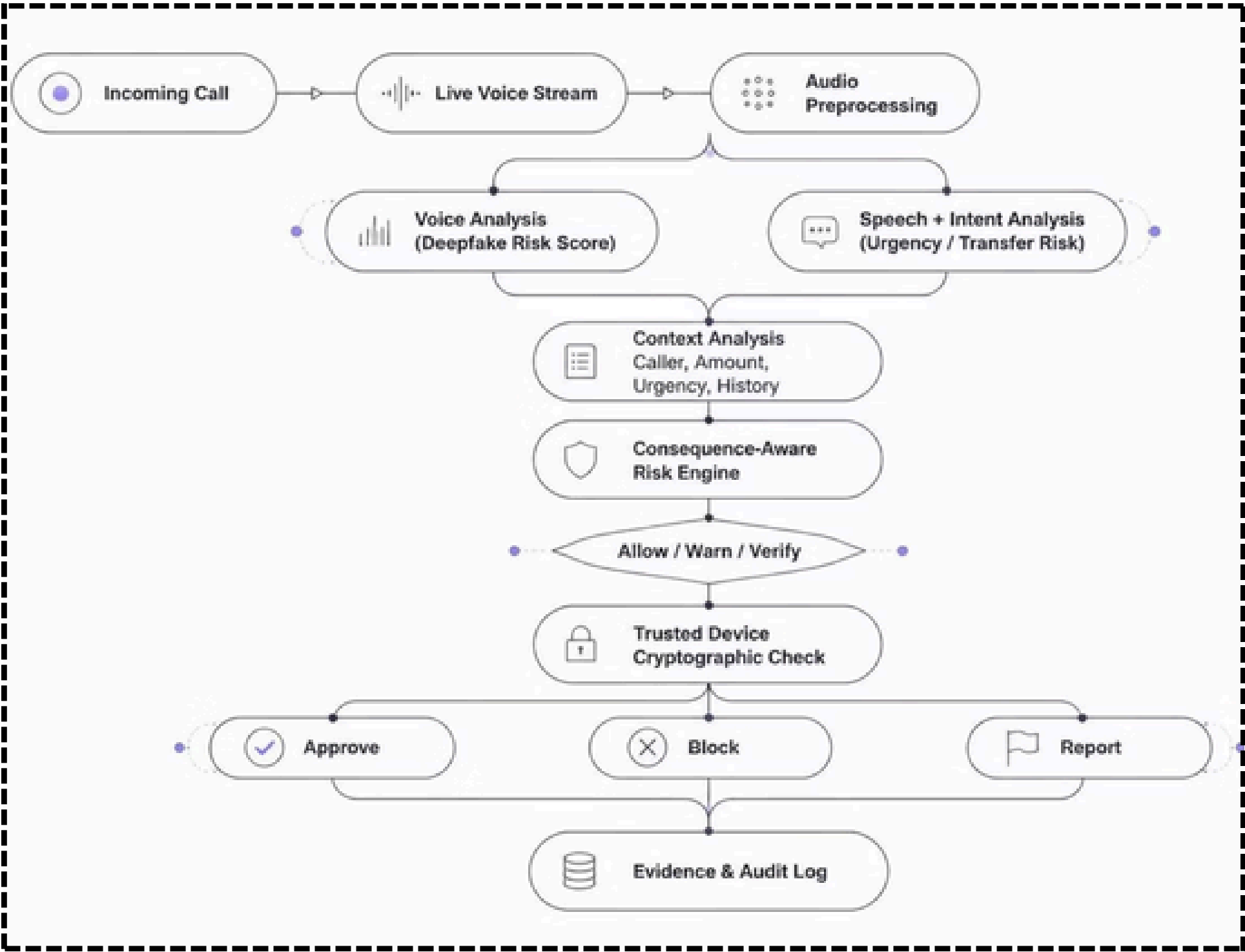
VOICE ≠ IDENTITY
Don't trust the voice. Verify the action.

RISK-AWARE VERIFICATION

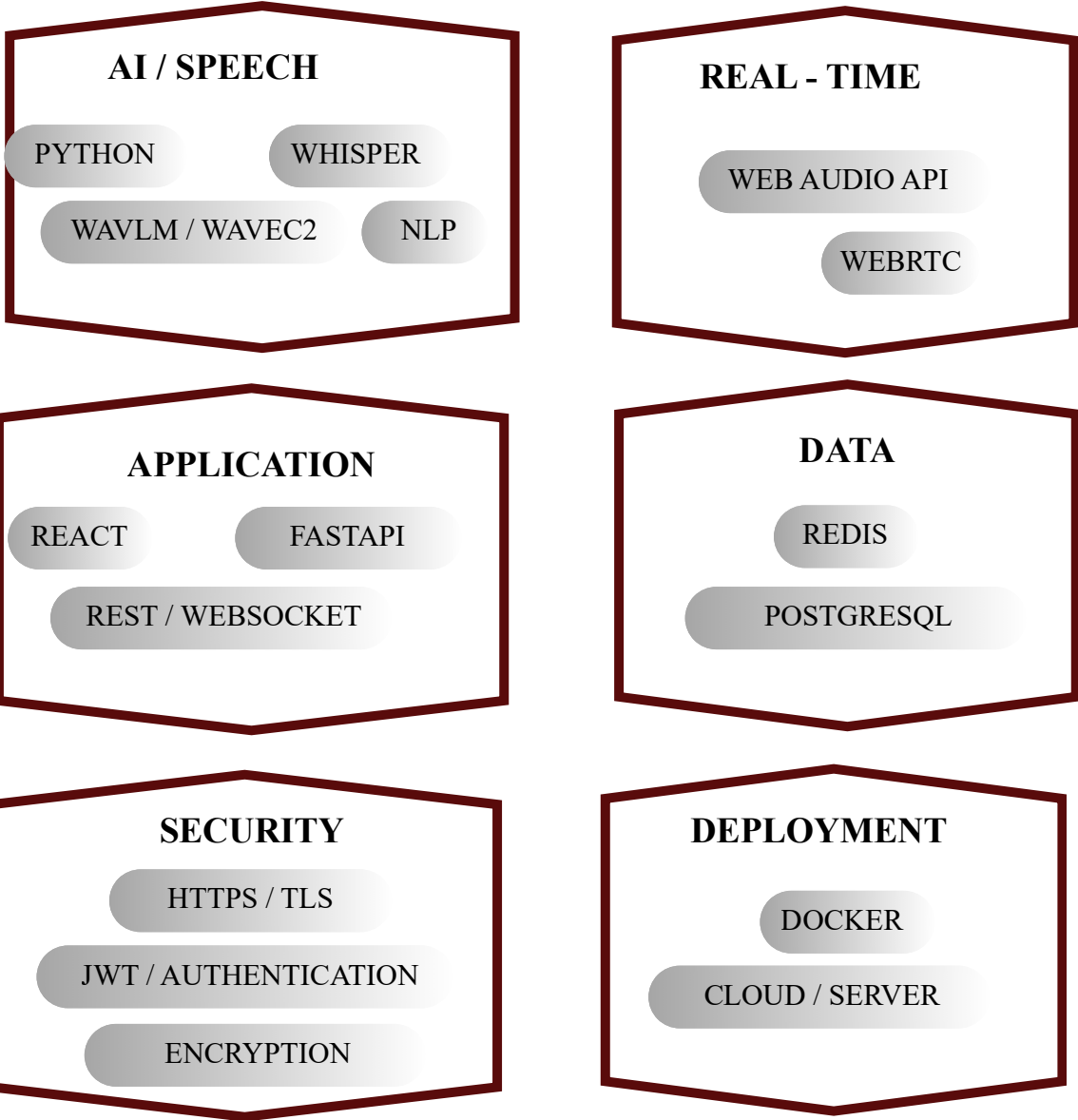
More risk → More verification.

DETECT → PREVENT
Detect the threat. Stop the action.

TRUSTBRIDGE SYSTEM ARCHITECTURE



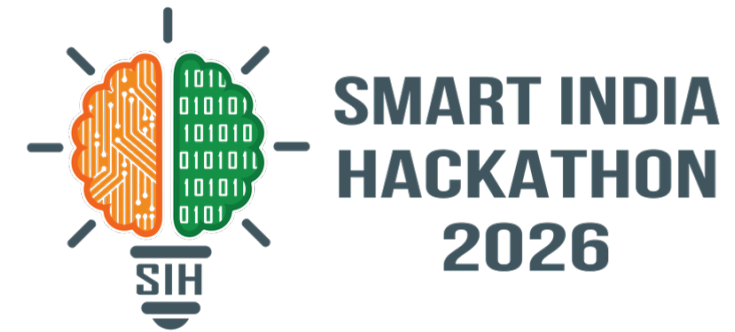
TECHNOLOGY STACK





FEASIBILITY AND VIABILITY

A practical path from AI prototype to real-world fraud prevention



FEASIBLE NOW

- Open deepfake & speech datasets
- Pretrained WavLM / Wav2Vec2 & ASR models
- Mature Python, PyTorch & FastAPI ecosystem
- Software-first WebRTC / VoIP MVP
- Standard cryptographic challenge-response
- Modular, cloud-ready architecture

KEY RISKS

- False positives and false negatives
- Unseen voice-cloning techniques
- Noise, compression & accent variation
- Trusted device unavailable / user timeout
- Voice privacy & consent compliance
- Cellular call integration limitations

MITIGATION STRATEGY

- Confidence thresholds: Allow / Warn / Verify
- Cross-dataset testing for generalization
- Noise & compression augmentation
- Trusted-device challenge-response
- Timeout → secondary verification → delay/block
- Consent, encryption & minimum data retention

VIABILITY & SCALE

FEASIBILITY ANALYSIS

- BUILDABLE:** AI models, cloud tools and WebRTC are available today.
- COST-VIABLE:** Start with selective risk analysis and a controlled pilot.
- SCALABLE:** API architecture supports financial institutions, telecom providers and government deployments.

DEPLOYMENT ROADMAP

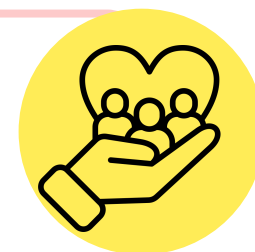
MVP → Pilot → Sector Integration → Scale

FAIL-SAFE FRAUD PREVENTION FLOW

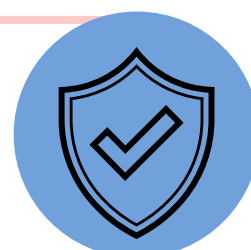


! “High-risk actions never silently fail open.”

PROTECTS PEOPLE
Reduces impersonation risk



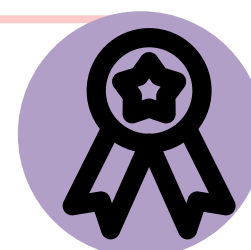
PREVENTS FRAUD
Stops loss before action



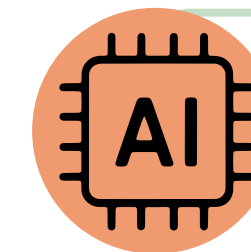
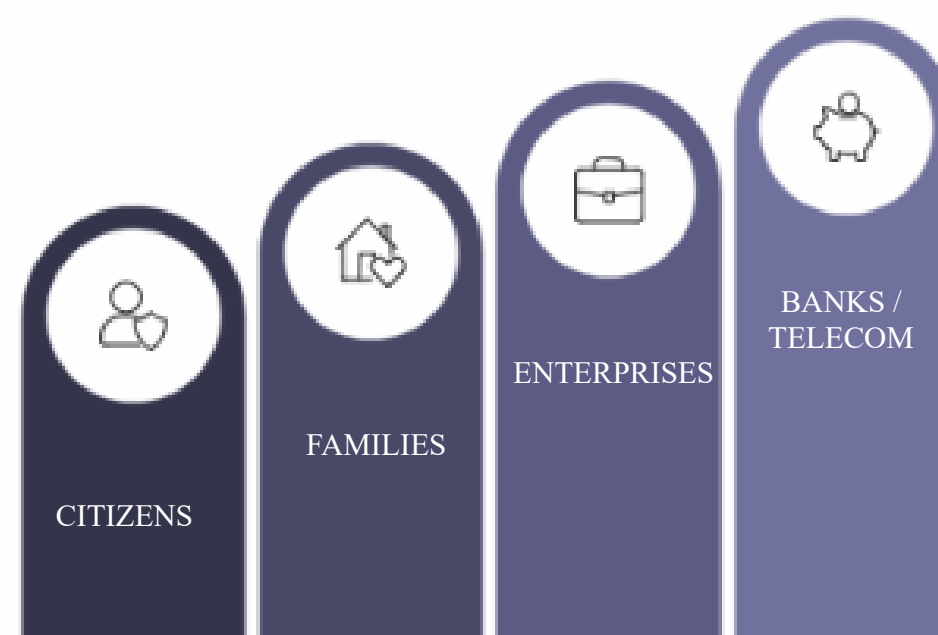
REDUCES MANIPULATION
Detects urgency and pressure



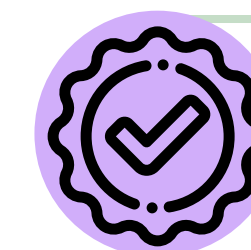
BUILDS TRUST
Safer voice communication



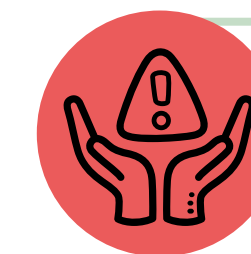
Who TrustBridge Protects ?



MULTI-SIGNAL AI
Voice + intent + context



INDEPENDENT VERIFY
Trusted-device approval



RISK-BASED ACTION
Allow • Warn • Block



AUDITABLE EVIDENCE
Report-ready decision logs

Suspicious Voice



AI Risk Analysis



Consequence Assessment



Independent Verification



Prevented Fraud



ACADEMIC FOUNDATION

- ASVspoof 2021 — Spoofed & Deepfake Speech Detection
- Audio Deepfake Detection Generalization — Müller et al.
- Deepfake Detection Survey & Domain Generalization
— Yi et al. / Xie et al.



DATASETS & BENCHMARKS

- ASVspoof 2019 / 2021
Standard synthetic-speech detection benchmark
- In-The-Wild Dataset
Real-world deepfake evaluation benchmark
- Used for training, fine-tuning, and robustness testing



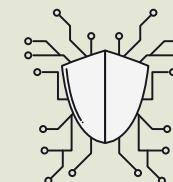
COMPETITOR LANDSCAPE

HIYA : Caller ID and Scam-Call Detection

PINDROP : Enterprise Voice, Call-risk, & Deepfake Detection

TRUSTBRIDGE : Deepfake risk + Intent + Context +
Independent Verification

NATIONAL CYBER ALIGNMENT



- National Cyber Crime Reporting Portal
cybercrime.gov.in
- 1930 Cyber Fraud Helpline
Fraud reporting and escalation reference
- Sanchar Saathi / Chakshu
Indian telecom fraud-reporting ecosystem

**“Detection alone is insufficient for high-consequence actions.
TrustBridge adds independent verification before authorization.”**